Event-Driven Sentiment Shifts Across Multiple Platforms: GPT Milestones and Social Media Discourse

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***Abstract*—This study analyzes public sentiment toward generative AI across YouTube, Weibo, and Bilibili, focusing on major GPT-related events. Using BERT-based sentiment models and time-series analysis, I quantify event-driven changes and compare cross-cultural responses, providing new insights into the impact of disruptive AI technologies.**

***Keywords*—Chatgpt NLP WebCrawler Bert**

I. Introduction

A. Background and Project Goal

The global rise of generative AI, especially ChatGPT, has sparked varied responses worldwide. However, systematic cross-platform analysis of ordinary user sentiment is lacking. This study compares reactions on American and Chinese social media, exploring how major AI events shape public opinion.

The goal is to quantitatively evaluate sentiment trends related to generative AI milestones across platforms by applying scalable NLP methods to large social media datasets. The project identifies key triggers and cultural differences in sentiment shifts

B. Literature review

Recent advances in natural language processing, especially BERT and its multilingual versions, have greatly improved sentiment detection in social media. Transformers outperform earlier rule-based and lexicon-driven approaches in contextual and emotional analysis (Devlin et al., 2018; Pang et al., 2002; Liu, 2012). However, most studies target a single platform or small regional data. Recent comparative work (Xu et al., 2022; Gao et al., 2021) highlights the difficulties of cross-cultural sentiment analysis, such as linguistic diversity and platform-specific challenges. While multi-lingual BERT has improved cross-language classification, further research is needed on event-driven and temporal patterns, especially for understanding public responses to generative AI.

II. Method

A. Data Source

All data utilized in this study were collected via web scraping techniques. YouTube comments were obtained using the official YouTube Data API, Bilibili data were retrieved through analysis of site JSON files, and Weibo comments were acquired with an open-source crawler from GitHub (https://github.com/zhouyi207/WeiBoCrawler.git). The crawling processes primarily targeted posts and videos ranked by the platforms' respective popularity algorithms, resulting in a dataset with minimal irrelevant information. Data preprocessing included removing duplicate comments, filtering out entries consisting exclusively of symbols, excluding comments shorter than five characters, eliminating advertisements, retaining only English and Chinese language comments, standardizing date formats to datetime, and stripping HTML tags. Specifically, I used the official YouTube v3 API to collect 200 comments each from 16 popular GPT-related videos. For Bilibili, I retrieved 200 comments from each of six trending GPT-related videos by extracting JSON data directly from the site. Weibo data comprised the top 1,200 posts on GPT topics from 2023 to 2025, gathered using the referenced GitHub crawler. Due to platform policy restrictions and legal considerations, I did not pursue further data collection from TikTok or X.

B. Data Preprocessing

This study incorporates the number of likes each comment received as a component of the overall weighting in sentiment score calculations, reflecting the collective endorsement of specific emotions by other users. Additionally, the NLP-generated confidence score for each comment serves as a secondary weight, representing the intensity of the expressed sentiment. All subsequent analyses are based on monthly averages of these weighted sentiment scores. The resulting formula is as follows:

$$\overline{MonthlyResult}_m = \frac{1}{N_m}\sum_{i=1}^{N_m}(label_i \times score_i \times \log(like_i + 1))$$

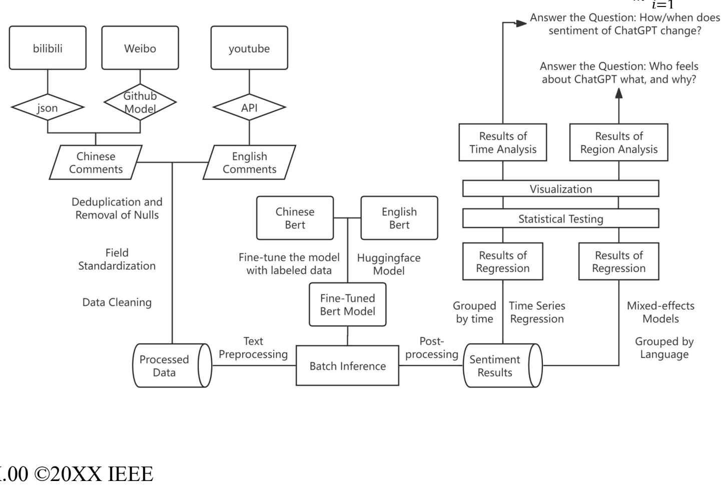

Fig. 1. Analytical Process Flowchart

C. NLP Model and Training

For NLP analysis, the project adopts mainstream BERT models and leverages the Transformers library in Python. Prior to application, fine-tuning is necessary for selected BERT variants. For Chinese-language sentiment classification, I chose the bert-base-chinese model and manually labeled 460 samples—of which 368 formed the training set and 92 the validation set. The entire training was conducted over five epochs, with each epoch involving 80 samples. Training loss decreased consistently as more data were included, whereas validation loss dropped initially but began to increase after the fourth epoch, indicating emerging overfitting. Consequently, I selected the model checkpoint at the third epoch as the final fine-tuned model. For English comments, I directly employed the published distilbert-base-uncased-finetuned-sst-2-english model from HuggingFace.

Epoch	Training Loss	Validation Loss
1	0.651800	0.691300
2	0.576900	0.714420
3	0.435100	0.643933
4	0.209100	0.752078
5	0.086900	0.840851

Fig. 2. Bert Training Process

D. Event Effect Analysis

To meet the project's main objective of examining how specific events affect public sentiment regarding GPT, I applied event effect analysis to the NLP-generated sentiment scores. In determining the pre- and post-event time windows, the study follows disciplinary conventions, using 12 months as the standard period for both sides. Six major events related to GPT were chosen for this analysis: the releases of GPT-4, Sora, GPT-4.5, GPT-5, Sora 2, and GPT's integration with Office 365.

$$\text{Before event:}\ \bar{Y}_{before} = \frac{1}{N_{\text{before}}}\sum_{i=1}^{N_{before}} Y_{before,i}$$

$$\text{After event:}\ \bar{Y}_{\text{after}} = \frac{1}{N_{\text{after}}}\sum_{i=1}^{N_{\text{after}}} Y_{after,i}$$

$$t = \frac{\bar{Y}_{before} - \bar{Y}_{after}}{\sqrt{\frac{s_{before}^2}{N_{before}} + \frac{s_{after}^2}{N_{after}}}}$$

E. Interrupted Time Series Analysis

Initial event effect analysis, including mean comparisons and t-tests, suggested that these methods alone were insufficient to capture nuanced shifts such as trend changes or sudden jumps in sentiment scores. Therefore, I conducted further ITSA (Interrupted Time Series Analysis) on YouTube monthly data to more comprehensively assess the dynamics of sentiment around these key events.

$$Y_t = \beta_0 + \beta_1 \cdot time + \beta_2 \cdot intervention_t + \beta_3 \cdot post_{e}vent_{t}ime + \epsilon_t$$

III. Result

A. Descriptive Analysis

I compared the weighted sentiment scores of comments across YouTube, Weibo, and Bilibili. Weibo displayed the most positive average sentiment, Bilibili was moderately positive, and YouTube was slightly negative on average. YouTube also had the largest variability in scores. The 25th and 75th percentiles show that most Weibo comments are positive, while Bilibili's are mostly neutral to positive. In contrast, YouTube had more negative or neutral comments, though positive sentiment is still present. These results suggest Weibo's user base is more positive, while YouTube shows greater negativity and fluctuation.

	youtube	weibo	bilibili
count	3090	1055	1073
mean	-0.464	2.169	0.721
std	3.584	1.862	2.309
min	-10.868	-6.196	-7.828
25%	-2.411	1.437	0
50%	0	2.102	0.515
75%	1.590	3.143	1.634
max	10.582	9.0278	10.091

Fig. 3. Descriptive Figure

B. Data Visualization by Platform

The following section presents platform-specific visualizations. In each chart, blue bars indicate comment volume, yellow lines represent monthly average weighted sentiment scores, and, to reduce noise, line graphs only display months with more than 30 comments. The red curve shows a smoothed trend line.

1) Bilibili: Due to the platform's popularity algorithm, Bilibili data are concentrated within just six months, resulting in fewer data points overall. Notably, in December 2023, OpenAI blocked a significant number of Chinese IPs and discontinued API services within mainland China, leading to a sharp decline in ordinary users' sentiment toward ChatGPT.

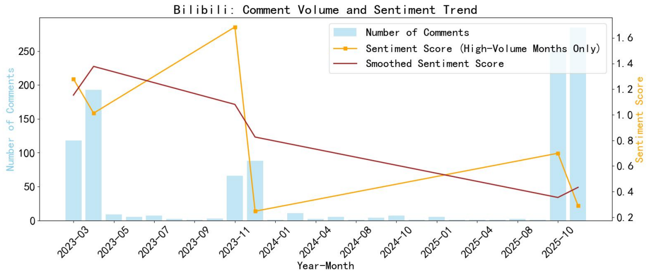

Fig. 4. Bilibili Comment Volume and Sentiment Trend

2) Weibo: Weibo data are more evenly distributed, and, given its role as a media and influencer-driven platform, the expressed sentiment tends to be more rational; moreover, positive attitudes are often incentivized for promotional purposes. Overall, following the release of GPT-4, sentiment on Chinese internet platforms was highly favorable. However, subsequent developments—such as the launch of Deepseek, underwhelming performance of GPT-5, and API bans—led to a sustained decrease in sentiment, reaching its lowest point last month (although it remained above zero).

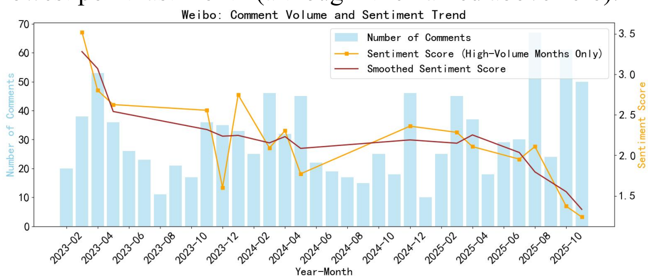

Fig. 5. Weibo Comment Volume and Sentiment Trend

3) Youtube: YouTube data are clustered around the upload months of selected videos but nonetheless provide useful insights into public attitudes and discourse. The overall sentiment is predominantly negative, mainly reflecting anxieties about the reliability of emerging technologies and fears related to job displacement. After the release of GPT-4, public sentiment remained relatively stable; however, it dropped sharply with subsequent launches of GPT-4.5 and GPT-5. Following the opening of GPT to adult content and the release of Sora 2, sentiment recovered to a higher level.

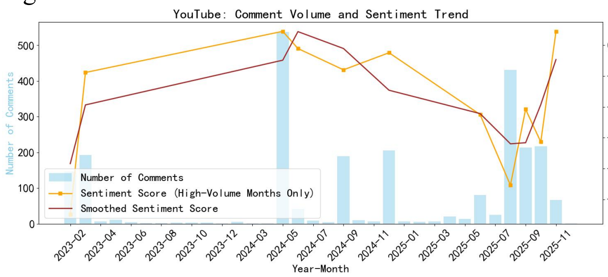

Fig. 6. Youtube Comment Volume and Sentiment Trend

C. *Event Effect Results*

After conducting event effect analyses for six key events across YouTube, Weibo, and Bilibili, the following table summarizes the mean sentiment scores before and after each event. In this visualization, green cells indicate that the event increased public sentiment on the corresponding platform significantly at the 5% significance level ($p < 0.05$), while red cells denote a significant decrease. White cells represent results that are not statistically significant — meaning the event did not have a measurable impact on public opinion from a statistical perspective.

Statistical results reveal that the releases of GPT-4 and Sora 2, as well as the integration of GPT with Office 365, significantly enhanced public sentiment toward GPT on YouTube. In contrast, the launch of GPT-4.5 led to a measurable decline in sentiment. On Weibo, most technological milestones resulted in decreased sentiment, with GPT-4.5' s release having no significant effect—likely because GPT-4.5 requires a paid, official account, limiting its accessibility to Chinese users. As a result, most people in China were not able to experience its features. For Bilibili, the data indicate that none of the selected events exhibited a strong relationship with public sentiment.

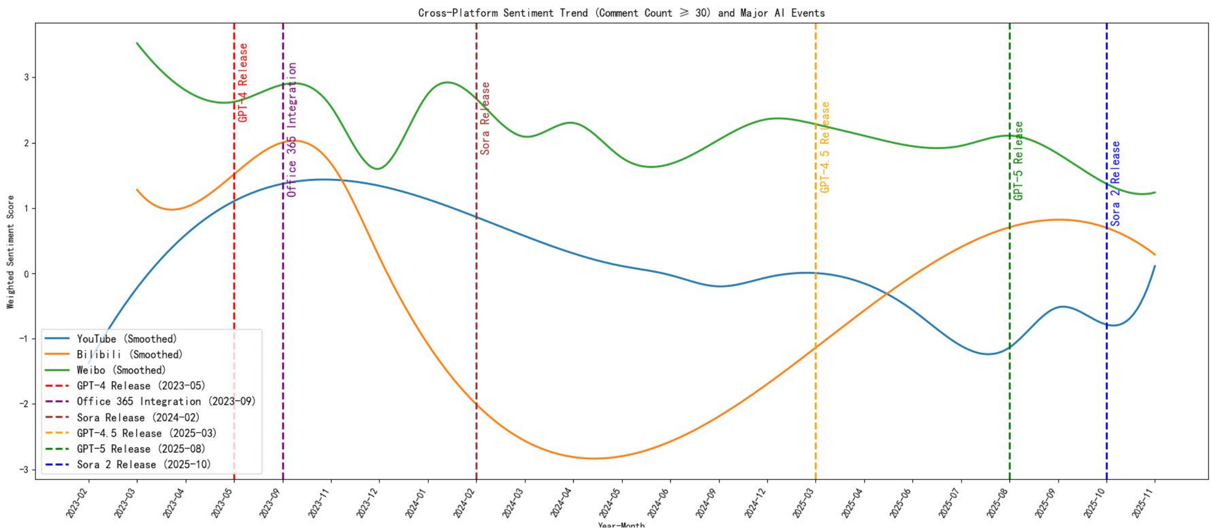

Fig. 7. Cross-Platform Sentiment Trend (Comment Count ≥ {count_threshold}) and Major AI Events

Event	GPT-4	Office Integration	Sora	GPT-4.5	GPT-5	Sora 2
YouTube	Mean(before): -0.648 Mean(after): 0.561 t: -2.222 p: 0.033	Mean(before): -0.653 Mean(after): 0.113 t: -2.844 p: 0.005	Mean(before): -0.15 Mean(after): 0.01 t: -0.787 p: 0.432	Mean(before): 0.013 Mean(after): -0.851 t: 5.825 p: 0.0	Mean(before): -0.336 Mean(after): -0.549 t: 1.109 p: 0.268	Mean(before): -0.747 Mean(after): 0.11 t: -2.91 p: 0.005
Bilibili	Mean(before): 1.114 Mean(after): 0.832 t: 1.27 p: 0.205	Mean(before): 1.118 Mean(after): 0.803 t: 1.476 p: 0.14	Mean(before): 1.036 Mean(after): 0.343 t: 2.823 p: 0.007	Mean(before): 0.247 Mean(after): 0.472 t: -0.958 p: 0.345	Mean(before): 0.131 Mean(after): 0.477 t: -1.237 p: 0.229	Mean(before): -0.296 Mean(after): 0.288 t: -1.674 p: 0.119
Weibo	Mean(before): 3.266 Mean(after): 2.29 t: 3.219 p: 0.002	Mean(before): 2.783 Mean(after): 2.101 t: 3.299 p: 0.001	Mean(before): 2.584 Mean(after): 2.096 t: 3.212 p: 0.001	Mean(before): 2.003 Mean(after): 1.757 t: 1.946 p: 0.052	Mean(before): 2.064 Mean(after): 1.427 t: 4.114 p: 0.0	Mean(before): 2.046 Mean(after): 1.238 t: 3.98 p: 0.0

Fig. 8. Event Effect Results

D. ITSA Results

ITSA results demonstrate that the integration of GPT with Office 365 produced a significant immediate increase in sentiment during the month of the event, but this was followed by a longer-term decrease. This suggests that the integration was initially well received and facilitated broader use of GPT, leading to positive reactions. However, over time, sentiment tapered off. The release of GPT-4.5, on the other hand, resulted in a significant decline in sentiment, indicating widespread dissatisfaction with this version among the public.

Event Month	Event Name	Coefficient intervention	p-value intervention	Coefficient post_time	p-value post_time
2023-05	GPT-4	0.192	0.7178	-0.043	0.1109
2023-09	Office 365 Integration	1.072	0.0434*	-0.074	0.014*
2024-02	Sora	-0.672	0.1585	0	
2025-03	GPT-4.5	-1.679	0.0184*	0.104	0.3869
2025-08	GPT-5	-0.924	0.3207	0.286	0.4296
2025-10	Sora 2	-0.424	0.6958	0.39	0.6315

Fig. 9. ITSA Results

IV. Conclusion

A. Major Findings

This study systematically examined public sentiment towards generative AI applications, focusing on critical events related to the development and deployment of large language models. By analyzing user comments from YouTube, Weibo, and Bilibili, several important insights emerged:

1) On YouTube, the releases of GPT-4 and Sora 2, as well as the integration of GPT with Office 365, significantly increased positive sentiment, while the launch of GPT-4.5 resulted in a measurable decline.

2) For Weibo, most technological milestones were associated with a downward trend in sentiment, with the exception of GPT-4.5, which did not significantly affect public opinion, likely due to limited accessibility in China.

3) Bilibili sentiment appeared relatively unaffected by these major events, indicating a weaker association between platform discourse and GPT-related milestones.

B. Limitations & Future Work

The study is limited by scope to three platforms and may not reflect broader global sentiment. Sample biases, platform restrictions, and event overlap may affect results. Future work could expand to other media, increase sample diversity, and assess multimodal sentiment for more comprehensive insights.

References

[1] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.

[2] Liu, B. (2012). Sentiment Analysis and Opinion Mining. Synthesis Lectures on Human Language Technologies, 5(1).

[3] Xu, Z., Li, X., & Zhang, H. (2022). Cross-platform sentiment analysis: A comparative study of public opinion on generative AI. Journal of Computational Social Science.

[4] Gao, F., Zhou, Y., & Sun, J. (2021). Multilingual and Cross-Platform Emotion Mining for Social Media. Information Processing & Management, 58(4).

[5] Zhou, Y., Sun, J., & Chen, Y. (2020). Robust Sentiment Detection in Large-scale Social Media Streams: Deep Language Modeling Approaches. ACM Transactions on Information Systems.